

AMATH 483 / 583 (Roche) - Homework Set 1

Due Wednesday April 9, 5pm PT

April 2, 2025

Homework 1 (85 points)

1. (+15) Integrate the following integrals. Show work.

(a) $\int_0^{\frac{\pi}{2}} \ln \sin(x) dx.$

(b) $\int_0^{\pi} \frac{x \sin(x)}{1 + \cos^2(x)} dx.$ Use $\int_0^a f(x) dx = \int_0^a f(a-x) dx.$

(c) $\int_0^{\frac{\pi}{2}} \frac{1}{1 + (\tan(x))^{\sqrt{2}}} dx.$

a.)

let $x = \frac{\pi}{2} - x$

$$A = \int_0^{\pi/2} \ln \left(\sin \left(\frac{\pi}{2} - x \right) \right) dx = \int_0^{\pi/2} \ln(\cos(x)) dx$$

$$2A = \int_0^{\pi/2} [\ln(\sin(x)) + \ln(\cos(x))] dx$$

$$= \int_0^{\pi/2} \ln(\sin(x) \cdot \cos(x)) dx$$

$$= \int_0^{\pi/2} \ln \left(\frac{1}{2} \sin(2x) \right) dx$$

$$= \int_0^{\pi/2} \ln \left(\frac{1}{2} \right) dx + \int_0^{\pi/2} \ln(\sin(2x)) dx$$

$$= \ln \left(\frac{1}{2} \right) x \Big|_0^{\pi/2} + \frac{1}{2} \int_0^{\pi} \ln(\sin(u)) du \quad \begin{matrix} u=2x & x=\frac{u}{2} \\ du=2 dx & u=2x=\pi \end{matrix}$$

$$2A = \ln \left(\frac{1}{2} \right) x \Big|_0^{\pi/2} + \frac{1}{2} \int_0^{\pi} \ln(\sin(u)) du$$

$$2A = \frac{\pi}{2} \ln \left(\frac{1}{2} \right) + A$$

$$A = \frac{\pi}{2} \ln \left(\frac{1}{2} \right)$$

$$= -\frac{\pi}{2} \ln(2)$$

$$\int_0^{\pi/2} \ln(\sin(x)) dx = -\frac{\pi}{2} \ln(2)$$

b) when $\int_0^a f(x) dx = \int_0^a \frac{x \sin(x)}{1 + \cos^2(x)} dx$

$$\int_0^a f(a-x) dx = \int_0^a \frac{(a-x) \sin(a-x)}{1 + \cos^2(a-x)} dx = \int_0^a \frac{(a-x) \sin(x)}{1 + \cos^2(x)} dx \quad \text{because } \begin{matrix} \sin(a-x) = \sin(x) \\ \cos(a-x) = \cos(x) \end{matrix}$$

$$\int_0^a \frac{x \sin(x)}{1 + \cos^2(x)} dx = \frac{1}{2} \left(\int_0^a \frac{x \sin(x)}{1 + \cos^2(x)} dx + \int_0^a \frac{(a-x) \sin(x)}{1 + \cos^2(x)} dx \right)$$

$$= \frac{1}{2} \int_0^a \frac{\pi \sin(x)}{1 + \cos^2(x)} dx$$

$$= \frac{\pi}{2} \int_0^a \frac{\sin(x)}{1 + \cos^2(x)} dx \quad u = \cos(x)$$

$$= \frac{\pi}{2} \int_0^a \frac{\sin(x)}{1 + u^2} \cdot \frac{du}{-\sin(x)} \quad du = -\sin(x) dx$$

$$= -\frac{\pi}{2} \int_0^a \frac{1}{1 + u^2} du$$

$$= -\frac{\pi}{2} \left[\tan^{-1}(u) \right]_0^a = -\frac{\pi}{2} \left[\tan^{-1}(\cos(x)) \right]_0^a$$

$$= -\frac{\pi}{2} \left[\tan^{-1}(-1) - \tan^{-1}(1) \right] = -\frac{\pi}{2} \left(-\frac{\pi}{4} - \frac{\pi}{4} \right)$$

$$= -\frac{\pi}{2} \left(-\frac{2\pi}{4} \right) = \frac{\pi^2}{4}$$

$$\frac{\pi^2}{4}$$

c) using $\int_0^a f(x) dx = \int_0^a f(a-x) dx$

$$\int_0^{\frac{\pi}{2}} \frac{1}{1 + (\tan(x))^{\sqrt{2}}} dx$$

$$= \int_0^{\frac{\pi}{2}} \frac{1}{1 + (\cot(x))^{\sqrt{2}}} dx$$

$$= \int_0^{\frac{\pi}{2}} \frac{1}{1 + \left(\frac{1}{\tan(x)} \right)^{\sqrt{2}}} dx$$

$$= \int_0^{\frac{\pi}{2}} \frac{1}{1 + (\tan(x))^{-\sqrt{2}}} dx$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{2}} \left(\frac{1}{1 + (\tan(x))^{\sqrt{2}}} + \frac{1}{1 + (\tan(x))^{-\sqrt{2}}} \right) dx$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{2}} \left(\frac{1}{1 + (\tan(x))^{\sqrt{2}}} + \frac{(\tan(x))^{\sqrt{2}}}{1 + (\tan(x))^{\sqrt{2}}} \right) dx$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{2}} 1 dx$$

$$= \frac{1}{2} \left[x \right]_0^{\pi/2}$$

$$= \frac{1}{2} \cdot \frac{\pi}{2} = \frac{\pi}{4}$$

$$\frac{\pi}{4}$$

2. (+15) Show work.

(a) Find $a, b \in \mathbb{R} \ni (1 + i\sqrt{3})^{11} = a + ib$.

(b) Find values of $(1 + i\sqrt{3})^{\frac{1}{5}}$.

(c) Solve for $w \in \mathbb{C}$ given $w^{\frac{4}{3}} + 2i = 0$.

a)

$$(1 + i\sqrt{3}) \xrightarrow{\text{Polar form}} 2 \left(\cos(\pi/3) + i \sin(\pi/3) \right)$$

$\sqrt{1+3} = 2$
 $\tan^{-1}(\sqrt{3}) = \pi/3$

$$\left[2 \left(\cos(\pi/3) + i \sin(\pi/3) \right) \right]^{11}, \text{ using De Moivre's Theorem}$$

$$= 2^{11} \left(\cos(11 \cdot \frac{\pi}{3}) + i \sin(11 \cdot \frac{\pi}{3}) \right)$$

$$= 2048 \left(\frac{1}{2} + i(-\sqrt{3}/2) \right)$$

$$= 1024 - 1024\sqrt{3}i$$

$$a = 1024$$

$$b = -1024\sqrt{3}$$

$$(1 + i\sqrt{3})^{11} = 1024 - 1024\sqrt{3}i = a + ib$$

b)

$$1 + i\sqrt{3} = 2e^{i\pi/3}$$

using De Moivre's Theorem

$$(1 + i\sqrt{3})^{1/5} = 2^{1/5} e^{i(\frac{\pi}{5} + \frac{2\pi k}{5})} \text{ when } k = 0, 1, 2, 3, 4$$

$$\begin{aligned} 1\text{st root: } 2^{1/5} e^{i\pi/5} &= 2^{1/5} \left(\cos\left(\frac{\pi}{5}\right) + i \sin\left(\frac{\pi}{5}\right) \right) \\ 2\text{nd root: } 2^{1/5} e^{i7\pi/5} &= 2^{1/5} \left(\cos\left(\frac{7\pi}{5}\right) + i \sin\left(\frac{7\pi}{5}\right) \right) \\ 3\text{rd root: } 2^{1/5} e^{i19\pi/5} &= 2^{1/5} \left(\cos\left(\frac{19\pi}{5}\right) + i \sin\left(\frac{19\pi}{5}\right) \right) \\ 4\text{th root: } 2^{1/5} e^{i13\pi/5} &= 2^{1/5} \left(\cos\left(\frac{13\pi}{5}\right) + i \sin\left(\frac{13\pi}{5}\right) \right) \\ 5\text{th root: } 2^{1/5} e^{i5\pi/5} &= 2^{1/5} \left(\cos\left(\frac{5\pi}{5}\right) + i \sin\left(\frac{5\pi}{5}\right) \right) \end{aligned}$$

c)

$$w^{4/3} + 2i = 0$$

$$w^{4/3} = -2i$$

using De Moivre's Theorem

$$r^{4/3} e^{i4\theta/3} = 2e^{i3\pi/2}$$

$$r^{4/3} = 2$$

$$r = 2^{3/4}$$

$$\frac{4\theta}{3} = \frac{3\pi}{2} + 2\pi k$$

$$\theta = \frac{9\pi}{8} + \frac{3\pi k}{2} \text{ when } k = 0, 1, 2, 3$$

$$\begin{aligned} 1\text{st root: } 2^{3/4} e^{i9\pi/8} &= 2^{3/4} \left(\cos\left(\frac{9\pi}{8}\right) + i \sin\left(\frac{9\pi}{8}\right) \right) \\ 2\text{nd root: } 2^{3/4} e^{i21\pi/8} &= 2^{3/4} \left(\cos\left(\frac{21\pi}{8}\right) + i \sin\left(\frac{21\pi}{8}\right) \right) \\ 3\text{rd root: } 2^{3/4} e^{i33\pi/8} &= 2^{3/4} \left(\cos\left(\frac{33\pi}{8}\right) + i \sin\left(\frac{33\pi}{8}\right) \right) \\ 4\text{th root: } 2^{3/4} e^{i45\pi/8} &= 2^{3/4} \left(\cos\left(\frac{45\pi}{8}\right) + i \sin\left(\frac{45\pi}{8}\right) \right) \end{aligned}$$

3. (+15) Write the following as the ratio of integers. Show work.

(a) $1 + 10^{-2} + 10^{-4} + 10^{-6} + \dots$

(b) $376.376376\dots$

(c) $.999\overline{9}$

a)
$$S = 1 + \frac{1}{100} + \frac{1}{100^2} + \frac{1}{100^3} + \dots$$

$$= \frac{1}{100} \left(100 + 1 + \frac{1}{100} + \frac{1}{100^2} + \frac{1}{100^3} + \dots \right)$$

$$S = \frac{1}{100} (100 + S)$$

$$100S = 100 + S$$

$$99S = 100$$

$$S = \frac{100}{99}$$

$$S = \frac{100}{99}$$

c)
$$z = 0.99999\dots$$

$$= \frac{9}{10} + \frac{9}{100} + \frac{9}{1000} + \dots$$

$$= \frac{1}{10} \left(9 + \frac{9}{10} + \frac{9}{100} + \dots \right)$$

$$10z = 9 + z$$

$$9z = 9$$

$$z = 1$$

$$z = 1 = \frac{1}{1}$$

b) $x = 376.376376\dots$

$$x = 376 + 0.376 + 0.000376 + \dots$$

$$= \frac{1}{1000} (376000 + 376 + 0.376 + \dots)$$

$$1000x = 376000 + x$$

$$999x = 376000$$

$$x = \frac{376000}{999}$$

$$x = \frac{376000}{999}$$

4. (+10) Estimate the following as the ratio of integers using the secant approximation. Show work.

(a) $(1.1)^{\frac{1}{3}}$

(b) $\sqrt{8.5}$

a) $f(x) = x^{1/3}$ at $x = 1.1$
 Secant approximation

$$f(x) \approx f(x_1) + \frac{f(x_2) - f(x_1)}{x_2 - x_1} \cdot (x - x_1)$$

$$f(1.1) \approx f(1) + \frac{f(1.2) - f(1)}{1.2 - 1} (1.1 - 1)$$

$$= 1 + \frac{(1.2)^{1/3} - 1}{0.2} (0.1)$$

$$\approx 1 + 0.0313 = \frac{1031}{1000}$$

$$(1.1)^{1/3} \approx \frac{1031}{1000}$$

b) $f(x) = x^{1/2}$ at $x = 8.5$

$$f(8.5) \approx f(8) + \frac{f(9) - f(8)}{9 - 8} (8.5 - 8)$$

$$= \sqrt{8} + \frac{1}{2} (3 - \sqrt{8})$$

$$= 2\sqrt{2} + \frac{3}{2} - \sqrt{2}$$

$$= \sqrt{2} + \frac{3}{2} \approx 2.914 = \frac{2914}{1000} = \frac{1457}{500}$$

$$\sqrt{8.5} \approx \frac{1457}{500}$$

5. (+10) Given $(x + y + z)^7$, find the expansion coefficients of the following terms. Show work.

(a) $x^2 y^2 z^3$

(b) $x^3 z^4$

a) using multinomial theorem

$$\frac{7!}{2!2!3!} = 210$$

$$\boxed{210}$$

b)

$$\frac{7!}{3!0!4!} = 35$$

$$\boxed{35}$$

6. (+5) Given $(x + 2y - 3z + 2w + 5)^{16}$, find the expansion coefficient of the following term. Show work.

(a) $x^2 y^3 z^2 w^5$

a)

$$\frac{16!}{2!3!2!5!4!} \cdot (1^2 \cdot 2^3 \cdot (-3)^2 \cdot 2^5 \cdot 5^4)$$

$$\boxed{\frac{16! \cdot 5^3}{6}}$$

$$= \frac{16! \cdot 5^3}{6}$$

7. (+10) Two numbers $a, b \in \mathbb{Z}$ are relatively prime when $\gcd(a, b) = 1$, or $\exists x, y \in \mathbb{Z}$ with $ax + by = 1$. Recall, $c = \gcd(a, b)$ when $c|a$ and $c|b$, and for any other divisor d of a, b then $d|c$.

(a) For any $n \in \mathbb{Z}^+$, prove $8n + 3$ and $5n + 2$ are relatively prime. Show work. Hint: try Euclid's algorithm

(b) Find the $\gcd(250, 111)$ and show result as linear combination of these integers. Show work.

a) $\gcd(8n + 3, 5n + 2) = \gcd(5n + 2, 8n + 3 - (5n + 2))$

$$= \gcd(5n + 2, 3n + 1)$$

$$\gcd(5n + 2, 3n + 1) = \gcd(3n + 1, 5n + 2 - (3n + 1))$$

$$= \gcd(3n + 1, 2n + 1)$$

$$\gcd(3n + 1, 2n + 1) = \gcd(2n + 1, 3n + 1 - (2n + 1))$$

$$= \gcd(2n + 1, n)$$

$$\gcd(2n + 1, n) = \gcd(n, 2n + 1 - 2n)$$

$$= \gcd(n, 1) = 1$$

$$\gcd(8n + 3, 5n + 2) = 1$$

This shows $8n + 3$ and $5n + 2$ are relatively prime for any $n \in \mathbb{Z}^+$.

b)

$$250 = 111 \cdot 2 + 28$$

$$1 = 28 - 27 \cdot 1$$

$$111 = 28 \cdot 3 + 27$$

$$1 = 28 - (111 - 28 \cdot 3)$$

$$28 = 27 \cdot 1 + 1$$

$$1 = 28 \cdot 4 - 111$$

$$27 = 27 \cdot 1 + 0$$

$$1 = 4(250 - 111 \cdot 2) - 111$$

$$\text{remainder} = 0$$

$$1 = 4 \cdot 250 - 111 \cdot 8 - 111$$

$$\gcd(250, 111) = 1$$

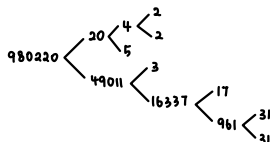
$$1 = 4 \cdot 250 - 9 \cdot 111$$

$$\gcd(250, 111) = 1$$

linear combination:

$$1 = 4 \cdot 250 - 9 \cdot 111$$

8. (+5) Write the prime factorization of 980220.



$$\boxed{980220 = 2^3 \cdot 3 \cdot 5 \cdot 17 \cdot 31^2}$$